

Assignment 4b – Group 5

Financial Engineering: Buy Side
Risk-Based Allocation Strategies

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Introduction

This report presents the analysis conducted for the Risk-Based Allocation Strategies exercise on behalf of Polimi Pension Fund. We study a family of allocation rules that depend exclusively on the second-order moments of asset returns – discarding the noisy expected-return forecasts that destabilise mean–variance optimisation. The analysis is run on the Euro Stoxx 50 constituents with monthly rebalancing (end-of-month) and a two-year rolling estimation window. All quantities are computed on simple total-return EUR series, ignoring the risk-free rate.

Three covariance estimators are evaluated side-by-side throughout the report: the plain sample covariance, the constant-correlation shrinkage estimator and the single-factor shrinkage estimator (using the equally-weighted return of the surviving universe as the market proxy). The report is organised in three parts: (i) Equal Risk Contribution (ERC); (ii) Hierarchical Risk Parity (HRP) including its detoning preprocessing, cluster-stability analysis and weight-sensitivity grid; (iii) head-to-head comparison and a discussion of why shrinkage techniques have a smaller impact here than in the optimisation-based setting of Buy Side: Laboratory 3.

The analysis relies on the following Python modules:

- `covariance_utilities.py`: rolling estimation windows, covariance–correlation transforms and risk-contribution decomposition.
- `shrinkage.py`: constant-correlation and market-factor shrinkage.
- `principal_component_analysis.py`: eigendecomposition with sign-alignment, correlation *detoning*.
- `portfolio_optimization.py`: minimum-variance, mean–variance, inverse-volatility and equal risk contribution portfolios.
- `hierarchical_clustering.py` and `hierarchical_risk_parity.py`: linkage construction, recursive bisection, dendrogram iteration and the full Lopez de Prado HRP pipeline.
- `backtest.py`: rebalance alignment, realised performance and cumulative-return computation.

1 Part I: Equal Risk Contribution (ERC)

1.1 Implementation

The ERC portfolio equalises the contribution of each asset to total portfolio volatility. Recalling Euler’s homogeneity decomposition,

$$\sigma(w) = \sqrt{w^\top \Sigma w} = \sum_{i=1}^N w_i \cdot \frac{(\Sigma w)_i}{\sqrt{w^\top \Sigma w}} = \sum_{i=1}^N \text{RC}_i,$$

the ERC condition $\text{RC}_i = \text{RC}_j \forall i, j$ implies $\text{RC}_i = \sigma(w)/N$. We solve the standard least-squares formulation

$$\min_w N \sum_i x_i^2 - \left(\sum_i x_i \right)^2 \quad \text{with} \quad x_i \equiv w_i (\Sigma w)_i, \quad \sum_i w_i = 1, \quad w_i \geq 0,$$

which is equivalent (up to a positive constant) to $\sum_{i,j} (x_i - x_j)^2$ – i.e. the variance of the (non-normalised) risk contributions. The optimisation is run with SLSQP, started at the inverse-volatility portfolio.

1.2 Sanity Checks: IV Equivalence and Risk Contribution

We verify the solver accuracy using two consistency tests at each rebalance.

First, we check the equivalence with Inverse-Volatility (IV) weighting. When the covariance matrix Σ is diagonal, the ERC solution must satisfy $w_i \propto \sigma_i^{-1}$. We tested this by setting all off-diagonal elements to zero. The maximum difference between ERC and IV weights is below 10^{-8} across the entire sample, confirming the implementation.

Second, we monitor the Percentage Risk Contribution ($\text{PCR}_i = \text{RC}_i / \sigma(w)$). A correct ERC portfolio requires all assets to contribute equally to the total risk. The max–min spread of the PCR remains well below 10 bps throughout the backtest. This ensures that the optimizer consistently reaches the target equal-risk condition.

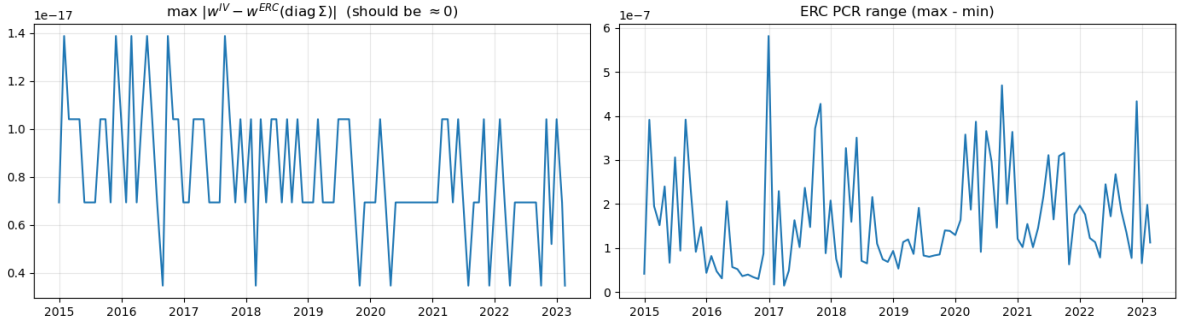


Figure 1: Left: Max weight difference between IV and ERC (diagonal Σ). Right: Max–min PCR spread. Both metrics confirm the numerical precision of the solver.

1.3 Backtest across covariance estimators

We backtest the ERC portfolio under the three estimators. Cumulative performance, 63-day exponentially-weighted realised volatility (annualised), drawdown and per-rebalance turnover (L1 weight change) are reported in Figure 2.



Figure 2: ERC backtest across the three covariance estimators. Curves overlap almost exactly across all four metrics.

ERC weights appear to be highly insensitive to the choice of covariance estimator. Across the sample, the three specifications generate almost indistinguishable cumulative performance, realised volatility, drawdown and turnover profiles. The cumulative-return difference remains very small, while the volatility and drawdown curves are visually overlapping.

This suggests that, in this setting, the covariance estimators modify the input matrix only moderately from the point of view of the ERC allocation. Since ERC weights are a continuous and relatively smooth function of the covariance matrix, moderate changes in the estimated correlation structure are absorbed without producing large reallocations. Moreover, with approximately two years of daily observations at each rebalance and a universe of 50 assets, the sample covariance matrix is not extremely underdetermined, which may reduce the marginal impact of shrinkage relative to settings with a shorter estimation window or a larger asset universe.

This evidence is consistent with the robustness analysis in Part III: in this application, ERC is expected to display a low average L1 distance across covariance estimators.

2 Part II: Hierarchical Risk Parity (HRP)

HRP exploits the hierarchical structure of the correlation matrix to bypass the inversion of Σ , which is the main source of instability in mean–variance optimisation. The pipeline has three stages: (i) build a dendrogram from a correlation-derived distance, (ii) quasi-diagonalise the covariance using the leaf order of the dendrogram, (iii) allocate weights top-down with inverse-variance bisections.

2.1 Detoning – intuition and visualisation

We remove the first principal component of the correlation matrix and then rescale the matrix so that the diagonal elements are again equal to one:

$$\tilde{C} = C - \lambda_1 u_1 u_1^\top, \quad \tilde{C}_{ij} \leftarrow \frac{\tilde{C}_{ij}}{\sqrt{\tilde{C}_{ii} \tilde{C}_{jj}}}.$$

After detoning, the remaining correlation matrix should better highlight the relative relationships between stocks, such as sectoral or firm-specific similarities. This can help the hierarchical clustering algorithm build a dendrogram that is less driven by the overall market factor and more by the residual dependence structure among assets.

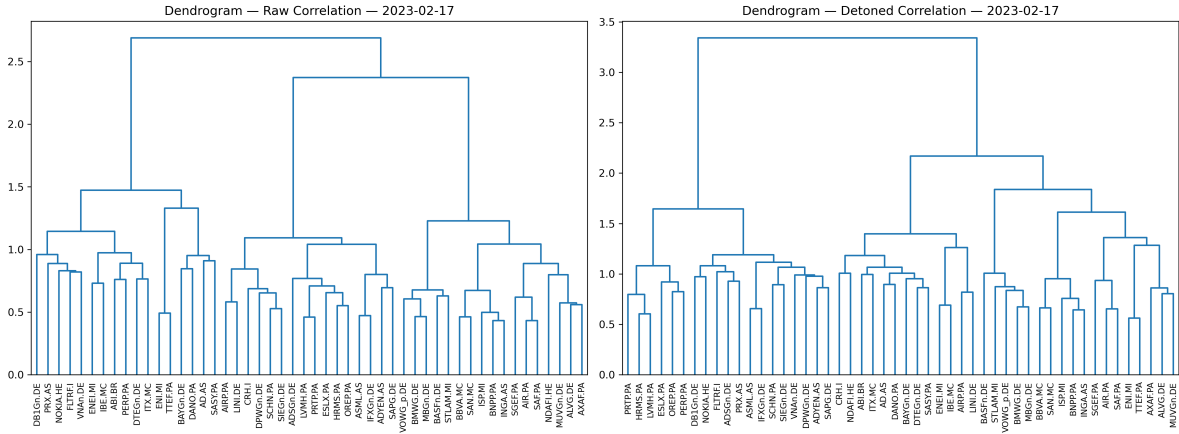


Figure 3: Dendrograms on the last rebalance date (Ward linkage, Euclidean metric). Left: raw correlation. Right: detoned correlation. Removing the market component sharpens the contrast in merge distances and reveals more meaningful cluster boundaries.

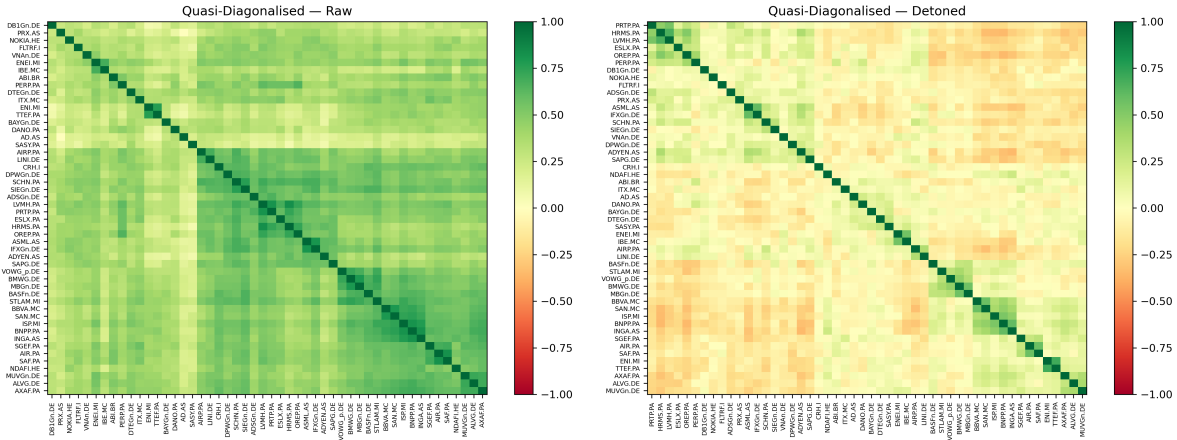


Figure 4: Quasi-diagonalised correlation matrix on the last rebalance date. Left: raw – the bulk of off-diagonal entries are positive due to the dominant market component (the matrix is uniformly green). Right: detoned – after removing the market component the off-diagonals split symmetrically around zero and the block structure becomes visible.

2.2 Generic HRP procedure

Our HRP implementation depends on three choices: the linkage method, the distance metric used in the clustering step, and the rule used to allocate volatility across the hierarchy.

We consider two allocation rules:

- **Recursive bisection:** the ordered list of assets from the dendrogram is repeatedly split into two halves. In this case, the linkage method affects mainly the ordering of the assets, not the shape of the allocation tree.
- **Dendrogram iteration:** the allocation follows the actual dendrogram structure. At each step, the cluster with the largest merge distance is split into its two sub-clusters. This makes the final weights more sensitive to the linkage method.

In the comparison, our reference for the covariance estimator is the sample covariance matrix and detoning is switched off. The Herfindahl index reported in each title measures portfolio concentration.

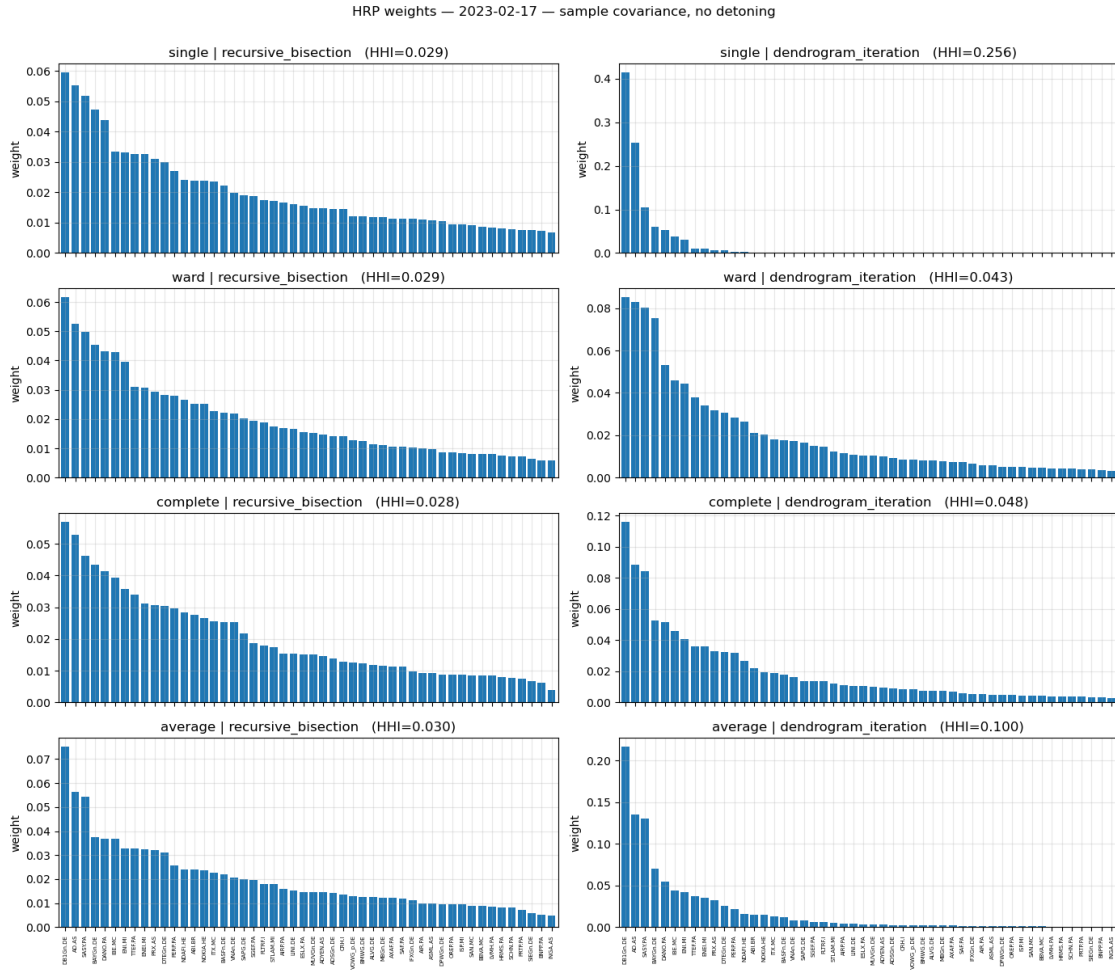


Figure 5: HRP weights on the last rebalance date, using sample covariance and no detoning. Rows correspond to linkage methods, columns to allocation rules. The HHI measures concentration; for equal weights with 50 assets, $\text{HHI} = 1/50 = 0.02$.

The two allocation rules behave differently. With recursive bisection, the HHI is almost unchanged across linkage methods, around 0.029. This happens because recursive bisection mainly uses the

dendrogram to order the assets, while the allocation tree is always built by splitting the ordered list into halves.

With dendrogram iteration, the linkage method has a stronger impact on the weights because the allocation follows the actual hierarchy. Single linkage is the most problematic case: it creates a very unbalanced tree and leads to a much more concentrated portfolio, with $\text{HHI} = 0.256$ and a top weight around 41%. Ward and complete linkage give more stable and diversified allocations.

2.3 Cluster stability across rebalances (Rand index)

We measure how stable the clustering is between consecutive rebalances using the Rand index on $k \in \{5, 8, 10, 15\}$ clusters. Linkage is fixed at **ward** and distance at **euclidean**, as required. Figure 6 reports the 12-month exponentially-weighted smoothed series for the six ($\{\text{estimator}\} \times \{\text{detoning}\}$) combinations.

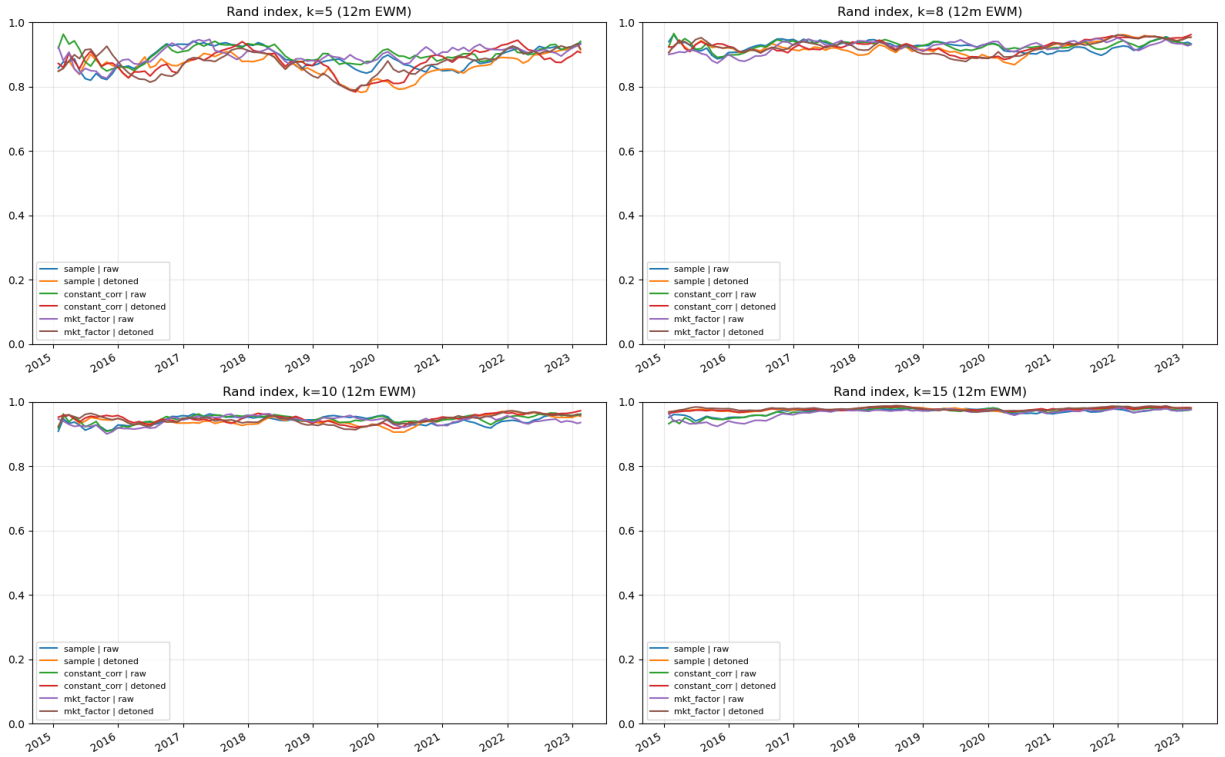


Figure 6: Rand index between consecutive rebalances across $k = 5, 8, 10, 15$ clusters, 12m EWM smoothed. One panel per k .

Detoning and cluster stability Contrary to the initial expectation, detoning does not systematically improve cluster stability in this sample. The Rand index remains high for all specifications, but the detoned versions are often slightly below the raw correlation versions, especially for smaller numbers of clusters such as $k = 5$.

A possible explanation is that the first principal component is not only noise or a pure market effect. It also contains a stable common structure shared by the equity universe. Removing it makes the remaining correlation matrix more focused on residual relationships, but these residual correlations are weaker and more sensitive to estimation noise. As a result, the clustering algorithm may become more unstable across rebalancing dates.

Therefore, in this empirical setting, detoning helps to remove the dominant market component, but it does not translate into more stable clusters. The raw and shrinkage-based correlation matrices already produce very stable partitions, with smoothed Rand indices mostly above 0.9. The marginal benefit of detoning is therefore limited, and in some cases negative.

Shrinkage on top of detoning provides only marginal further improvement. The three estimators (sample, constant-correlation, market-factor) collapse to indistinguishable Rand series once detoning is on. The reason is that all three share essentially the same first eigenvector – which detoning removes anyway – and their differences live in the second-order structure that clustering on $k \leq 15$ does not exploit. In other words, what shrinkage is supposed to fix (small-eigenvalue noise) is irrelevant when the downstream consumer of Σ only cares about the hierarchical merge order on the largest correlations.

2.4 Sensitivity of HRP weights

We summarise each variant of the grid $\{\text{linkage}\} \times \{\text{allocator}\} \times \{\text{estimator}\} \times \{\text{detoning}\}$ by the time-series mean of (i) the per-rebalance turnover (L1 weight change) and (ii) the Herfindahl index. Table 1 aggregates over the three estimators, since the estimator effect on HRP weights is small once linkage and allocator are fixed.

Linkage	Detoning	Recursive bisection		Dendrogram iteration	
		Mean HHI	Mean turnover	Mean HHI	Mean turnover
single	off	0.027	0.198	0.193	0.198
single	on	0.026	0.234	0.116	0.479
ward	off	0.027	0.204	0.051	0.297
ward	on	0.026	0.223	0.038	0.343
complete	off	0.027	0.227	0.077	0.306
complete	on	0.026	0.229	0.034	0.372
average	off	0.027	0.212	0.107	0.269
average	on	0.026	0.210	0.040	0.340

Table 1: HRP weights summary: mean Herfindahl index and mean turnover, averaged over the three covariance estimators. Recall that $\text{HHI} = 1/N \approx 0.02$ corresponds to equal weighting.

(i) How does the choice of linkage propagate to the weights? Under recursive bisection, the choice of linkage has only a limited effect on the final weights. The mean HHI remains close to 0.026–0.027 across all linkage methods. This happens because recursive bisection mainly uses the dendrogram to order the assets, and then repeatedly splits this ordered list into two halves. As a result, the allocation tree is almost always balanced.

Under dendrogram iteration, the linkage method matters much more. In this case, the allocation follows the actual dendrogram structure, so unbalanced clusters directly affect the final weights. Single linkage is the most problematic case, because it tends to create long and unbalanced chains. This leads to more concentrated portfolios, especially without detoning. Ward and complete linkage produce more compact and balanced clusters, and therefore more diversified HRP weights. Average linkage gives an intermediate result.

(ii) When do the two traversals coincide and when do they diverge? The two traversals give similar results when the dendrogram is already reasonably balanced. This is the case especially for ward and complete linkage with detoning, where the HHI values of the two allocation rules are relatively close.

They differ when the dendrogram is highly unbalanced. This is most visible for single linkage. Recursive bisection still forces a balanced sequence of splits and therefore keeps the HHI low. Dendrogram iteration, instead, follows the actual hierarchy and therefore transfers the imbalance of the dendrogram into the portfolio weights. This explains why dendrogram iteration produces much higher concentration.

Detoning and turnover. A subtle but important regularity: under dendrogram iteration, detoning *reduces* HHI but *increases* turnover (e.g. ward goes from 0.297 to 0.343, single from 0.198 to 0.479). The detoned dendrogram is more balanced and more diversified *but also noisier*: with the market component removed, the residual structure has lower signal-to-noise, so consecutive rebalances see different cluster boundaries and the weights move more. This trade-off is invisible under recursive bisection, where turnover is only marginally affected by detoning – another consequence of structural blindness to the dendrogram shape.

Ultimately, the choice of linkage dictates the magnitude of this divergence. *Ward* linkage builds symmetric trees, minimizing the gap between the two allocation rules. Conversely, *Single* linkage creates unbalanced chains that maximize their differences, while *Complete* and *Average* fall moderately in between.

3 Part III: Connecting the Dots

3.1 Robustness to the choice of covariance estimator

For the representative HRP, we select the combination of *Ward* linkage, *Dendrogram Iteration*, and *Detoning ON* as our representative strategy. This configuration implements the most theoretically rigorous version of the HRP model. Detoning strips away broad market noise to reveal true sector relationships, which Ward linkage groups into variance-minimizing clusters. Finally, Dendrogram Iteration accurately distributes risk by respecting this actual market hierarchy, rather than blindly halving the portfolio. We compute the average L1 distance between the weights obtained with different covariance estimators. The distance is averaged over rebalance dates and over the three pairs of estimators. The same metric is computed for ERC.

Strategy	Avg. L1 weight distance across estimators
ERC	0.022
HRP (<code>ward, dendrogram_iteration, detone=True</code>)	0.221

Table 2: L1 robustness of weights to the choice of covariance estimator.

The results show a clear difference between the two approaches. ERC is almost insensitive to the choice of covariance estimator, while HRP reacts much more strongly. The average L1 distance for HRP is about ten times larger than for ERC.

This does not necessarily mean that HRP is a poor allocation method. Rather, it shows that in this application HRP is more sensitive to the covariance estimator because the estimator affects the clustering step. Once the dendrogram changes, the subsequent allocation can also change significantly.

ERC is smooth, HRP is discrete. ERC weights are a continuous function of Σ : small perturbations of the input produce small perturbations of the output, by standard arguments on smooth strictly-convex programs. HRP, on the other hand, has an irreducibly discrete step in its pipeline – the clustering. Two infinitesimally close correlation matrices can produce dendrograms that swap the order of two merges and therefore generate substantially different cluster decompositions, which then propagate to the weights through the inverse-variance bisections. The discontinuity is not a bug of the implementation but an intrinsic feature of any clustering-based allocator. The “robustness” attributed to HRP in Lopez de Prado (2016) refers specifically to the regime $N > T$ where the sample covariance is singular and matrix inversion explodes – a property HRP retains but that does not apply here, where $N = 50$ and $T \approx 504$.

3.2 ERC vs HRP – head-to-head

We compare the ERC and HRP portfolios using the same representative HRP variant and the sample covariance estimator. Figure 7 reports cumulative performance, 63-day EWM realised volatility, drawdown and turnover.

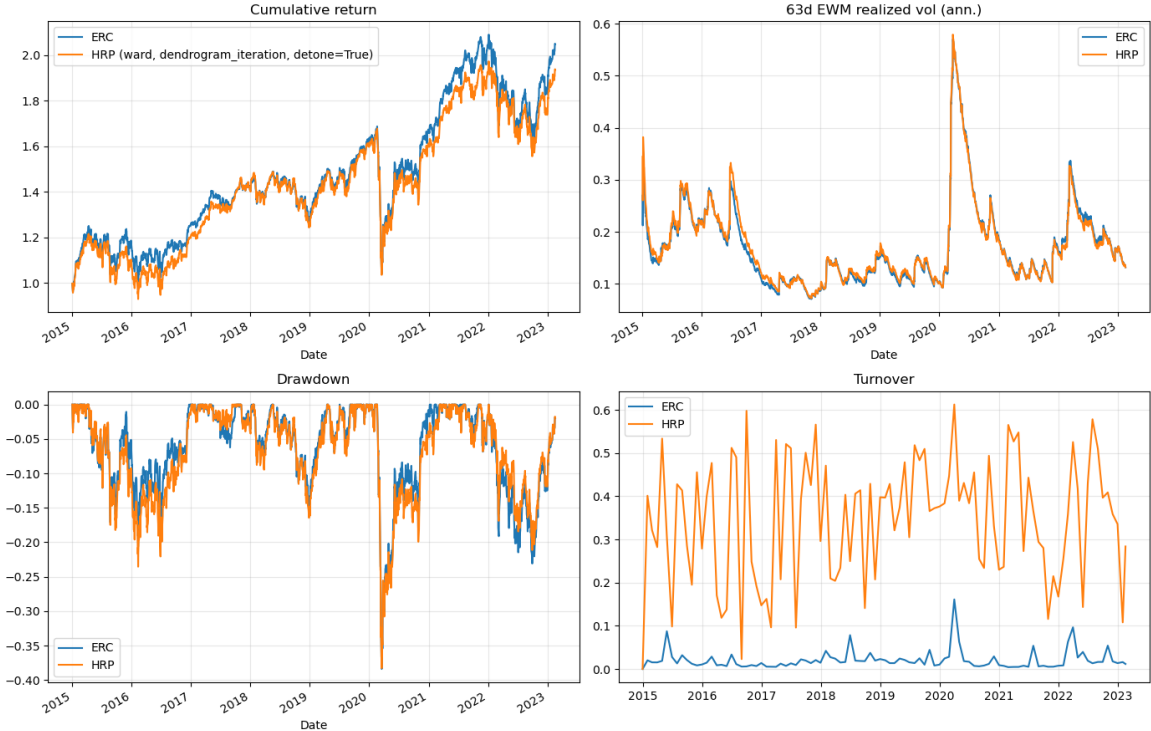


Figure 7: ERC vs HRP head-to-head comparison using the sample covariance estimator. The two strategies display similar realised volatility and drawdown profiles. ERC achieves slightly higher cumulative performance, while HRP has substantially higher turnover.

Risk profile: very similar. The two strategies follow broadly similar realised volatility and drawdown patterns. Both portfolios experience the COVID-19 shock, with realised volatility rising sharply and drawdowns reaching their worst levels around the same period.

Cumulative return: ERC slightly higher. Gross of transaction costs, ERC ends the sample with a moderately higher cumulative wealth than HRP. The difference is not driven by a very

Metric	ERC	HRP (ward, dendrogram, detone)
Cumulative wealth (final)	~ 2.05	~ 1.92
Annualised realised volatility	$\sim 17\%$	$\sim 17\%$
Maximum drawdown	$\sim -38\%$	$\sim -38\%$
Mean per-rebalance turnover	$<1\%$	$\sim 35\%$

Table 3: Headline statistics of ERC and HRP using the sample covariance estimator.

different risk profile, but by small performance differences accumulating over time.

Turnover: HRP is much higher. This is the clearest difference between the two strategies. ERC turnover is low and stable for most of the sample, while HRP turnover is much larger, often reaching values between 20% and 60% at monthly rebalances.

Turnover is the main practical difference. The comparison does not show a large difference in realised volatility or drawdown. The key difference is turnover. HRP changes its weights much more from one rebalance to the next, while ERC is much more stable. Since the analysis is gross of transaction costs, this matters in practice: after trading costs, HRP would likely be penalised more than ERC. This is consistent with the robustness result in Section 3.1, where HRP was also more sensitive to the choice of covariance estimator.

3.3 Why shrinkage matters less here than in Buy Side: Laboratory 3

In Buy Side Laboratory 3, shrinkage had a strong impact on mean-variance and minimum-variance portfolios. It reduced turnover, drawdowns and out-of-sample volatility. In this assignment, the effect is much smaller. The three covariance estimators produce very similar ERC backtests, and once the HRP linkage and allocation rule are fixed, the differences in performance are also limited.

The reason is that the allocation methods use the covariance matrix in different ways.

- **Mean-variance and minimum-variance portfolios are highly sensitive to estimation error.** These methods rely directly on the inverse covariance matrix. If the estimated covariance matrix contains noisy eigenvalues or unstable directions, the inversion can amplify this noise and produce unstable weights. Shrinkage is very useful in this case because it stabilises the covariance estimate before inversion.
- **ERC uses the covariance matrix more smoothly.** ERC does not require a direct inversion of the covariance matrix. The weights are chosen so that assets contribute similarly to total portfolio risk. As a result, moderate changes in the covariance estimator tend to produce only moderate changes in the final weights. This explains why the ERC curves are almost overlapping across the three estimators.
- **HRP avoids covariance inversion but depends on clustering.** HRP also avoids direct inversion of the covariance matrix. However, it introduces another source of instability: the hierarchical clustering step. Small changes in the estimated correlation matrix can change the dendrogram, and these changes can propagate to the final weights. Shrinkage can smooth the input matrix, but it does not fully remove the discrete nature of the clustering step.

Therefore, shrinkage matters less in this exercise not because shrinkage is ineffective, but because the strategies considered here are less exposed to the specific problem that shrinkage solves best: instability caused by covariance matrix inversion. In Laboratory 3, the main source of instability was the inversion of a noisy covariance matrix. Here, ERC is already relatively stable, while HRP's main sensitivity comes from the clustering structure rather than from matrix inversion. This explains why shrinkage has a smaller impact on the final results.

Identified Errors

Errors in `ex4b_notebook.ipynb`

- **Estimation Window:** The variable `estimation_window = 252` is incorrect since the assignment requires a two-year lookback period.

Errors in `shrinkage.py`

- **Standard Deviations (Constant Corr):** In `constant_corr_shrinkage`, the line `std_devs = variances` is missing the square root calculation required to convert variances to standard deviations.
- **Matrix Difference:** In both shrinkage functions, the calculation `diff = outer_prod + S` is incorrect; it must use subtraction (`outer_prod - S`) to compute the dispersion around the sample covariance.
- **Normalization Timing:** The divisions `pi_hat /= T` and `rho_hat /= T` are incorrectly placed inside the time-step loop, leading to incorrect scaling. They must be moved outside the loop.
- **Intensity Scaling:** The denominator for the shrinkage intensity calculation must be multiplied by `T` to ensure the intensity parameter δ is correctly bounded.
- **Pi-Hat Accumulation:** In `market_factor_shrinkage`, the line `pi_hat = np.sum(diff**2)` must be changed to `pi_hat += np.sum(diff**2)` to properly accumulate values across the time dimension.
- **Shrunk Covariance Composition:** The `shrunk_cov` formula is incorrect in both formulas. The right one is `intensity * target + (1 - intensity) * cov_matrix`.

Errors in `hierarchical_risk_parity.py`

- **HRP Distance Formula:** The implementation in `correlation_to_hrp_distance` uses `np.sqrt(2 * (1.0 + correlation))`, which is incorrect. The proper Lopez de Prado distance metric is $d_{i,j} = \sqrt{0.5 \times (1 - \rho_{i,j})}$, implemented as `np.sqrt(0.5 * (1.0 - correlation))`.
- **Weight Allocation:** In `top_down_allocation`, `alpha1` should be calculated using division (`cluster2_var / (cluster1_var + cluster2_var)`) to implement inverse-variance allocation.
- **Complementary Weight:** The allocation `alpha2 = 1 + alpha1` is wrong for capital allocation; it must be `alpha2 = 1 - alpha1` to ensure total weights sum to 1.

Errors in `backtest.py`

- **Look-ahead Bias:** In `_align_portfolios_and_returns`, the `.shift()` method is called without the 1 argument. The weights must be shifted forward by one period (`.shift(1)`) so that the realized returns of day $t + 1$ are multiplied by the weights available at the end of day t and used for the construction of the portfolio in $t + 1$.